

Damian Szczykutowicz

Piscataway, NJ 08854 | dls421@scarletmail.rutgers.edu | 908-416-3130 |
linkedin.com/in/damian-szczykutowicz | damianszczykutowicz.github.io/Portfolio

EDUCATION

RUTGERS UNIVERSITY, HONORS COLLEGE

Bachelor of Science, Mechanical Engineering | Minor in Computer Science

New Brunswick, NJ

Expected May 2028

GPA: 3.7/4.0 | Honors: Dean's Scholarship, Dean's Promise Scholarship, William Gawor Scholarship

Relevant Coursework: Thermodynamics, Fluid Mechanics, Heat Transfer, Mechatronics, Design of Mechanical Components, Dynamic Systems and Controls

EXPERIENCE

NOVARTIS - Manufacturing Engineering Intern

East Hanover, NJ

Radioligand Therapy Manufacturing (GxP)

May 2026 – Aug 2026

- Executed equipment lifecycle management for GxP radioligand production assets in BMRAM and Kneat, keeping 34 critical pieces of equipment qualified and inspection-ready
- Reviewed equipment history, calibration schedules, and maintenance records each qualification period to verify compliance and flag overdue actions or performance drift before they impacted production
- Executed periodic review protocols in Kneat, tracing each asset's qualified state against GxP requirements and documenting deviations for review
- Partnered with production, quality, MS&T, and process engineering to align documentation control and resolve record discrepancies across radioligand therapy manufacturing

COLDTRACK - Process Engineering Co-op

Edison, NJ

Data-Driven Process & Yield Improvement

Sep 2024 – May 2026

- Analyzed large-scale operational datasets in MATLAB, Python, and Excel to identify failure trends and root causes of yield loss and process inefficiency
- Applied Lean Six Sigma root cause analysis to eliminate process bottlenecks and increase throughput
- Built a data-driven cartonization model optimizing box selection, cutting packaging and logistics waste ~15% (~\$9M annual savings)

LEADERSHIP & ACTIVITIES

- Learning Assistant, Modern Physics, Rutgers University: completed Peer Leader Training; guide undergraduates through problem-solving and concept explanation

ENGINEERING PROJECTS

- **Autonomous Delivery UAV (AeroDrop)** - Designed a sub-\$200 UAV in SolidWorks with a topology-optimized 3D-printed airframe; validated structure and a servo-actuated payload-release mechanism via SolidWorks FEA; ran the full build from CAD through fabrication to validated flight testing
- **Object-Tracking Drone (Jetson + Vision)** - Integrated an NVIDIA Jetson, Pixhawk controller, OAK-D stereo vision, live telemetry, and power distribution on a carbon-composite frame, a distributed sense/compute/power system with real-time monitoring across subsystems
- **Long-Range VTOL Quadplane** - Designed a fixed-wing VTOL in CAD; used SolidWorks Flow Simulation and strength-to-weight optimization on the 3D-printed structure; validated with a successful hover-to-cruise transition flight

SKILLS

- **Systems & Reliability:** Equipment Qualification / Validation / Periodic Review, Reliability & Trending Analysis, Root Cause Analysis, Lean Six Sigma, Documentation Control, GxP, GD&T
- **Thermal & Mechanical Design:** SolidWorks (FEA, Flow Simulation / CFD, PDM), AutoCAD, OnShape, CadQuery, 3D Printing / Additive, Machining, Rapid Prototyping
- **Data & AI:** Python, MATLAB, SQL, R, Excel, AI-assisted engineering workflows
- **Systems Integration & Comms:** Embedded systems, telemetry (MAVLink), power distribution, FCC Amateur Radio (Ham) license; comfortable in IT-centric / Linux environments